

BIOMEDICAL ONCOLOGY APPLICATION OF GENE CO-EXPRESSION NETWORKS IN GLIOBLASTOMA MULTIFORME

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ABSTRACT. Conventional genomic analyses of cancer focus on individual high-risk mutations, yet many malignancies arise in their absence, underscoring the need for a more holistic screening paradigm. We propose graph networks as such a framework: genes are modeled as nodes, statistically significant co-expression relationships (Pearson’s correlation with p -value < 0.05 , Benjamini–Hochberg corrected) as weighted edges, and the resulting gene co-expression network encodes collective regulatory behavior invisible to single-gene methods. Using glioblastoma multiforme (GBM) and normal brain expression data, we identify important graph-theoretic signatures that distinguish cancer from normal networks. Each signature was tested for statistical significance via the Mann–Whitney U test and bootstrapped goodness-of-fit procedures. Our results confirm significant topological differences across all metrics evaluated, consistent with GBM manifesting as a network-level disorder rather than a collection of isolated mutations. These signatures lay the groundwork for a graph-theoretic biomarker framework applicable to early cancer detection and, prospectively, to a meta-analysis spanning multiple cancer types.

1. INTRODUCTION

Glioblastoma multiforme (GBM) is the most aggressive form of astrocytic tumor, characterized by rapid proliferation, extensive infiltration, and poor response to therapy. Despite decades of sequencing efforts, median patient survival after diagnosis remains under 15 months, and targeted single-gene therapies have achieved only modest gains [1]. This persistent failure highlights a fundamental limitation of reductionist approaches: cancer is not simply the consequence of one or two mutant genes, but an emergent phenomenon arising from dysregulated interactions across entire gene regulatory networks.

Graph theory provides a natural mathematical language for capturing such network-level phenomena. Representing genes as nodes and co-expression relationships as edges transforms a high-dimensional transcriptomic dataset into a topological object whose global properties can be analyzed with precision. Differences between healthy and malignant tissue can then be framed as differences in network topology — differences that may be quantified, tested statistically, and used as predictive signatures of malignancy.

This paper investigates whether GBM leaves reproducible graph-theoretic signatures in gene co-expression networks and whether such signatures can be rigorously characterized using tools from graph theory and spectral analysis.

2. APPLICATION OF GRAPH NETWORKS

Mathematical and computational models are increasingly indispensable in biomedical research, where in vivo and in vitro studies face both scientific and ethical constraints. In this study, all analyses were conducted in silico using glioblastoma multiforme and normal brain gene expression data. After systematic investigation, four graph-theoretic signatures were identified as particularly informative for distinguishing cancer from normal gene co-expression networks. Each captures a distinct structural property of the network, and together they provide a multi-dimensional characterization of malignancy.

2010 *Mathematics Subject Classification.* 05C69.

Key words and phrases. Gene co-expression, Pearson’s Correlation Coefficient, Mann–Whitney U test.

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